## 📋 Problem Statement — Customer Shopping Behaviour Analysis

**Business Problem**

A retail company wants to understand its customers' shopping behavior to improve sales, customer satisfaction, and long-term loyalty. The management has noticed changes in purchasing patterns across demographics, product categories, and seasons but lacks data-driven insights to act on them.

**Business Objective**

*"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

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**Tools Used**

- **Python** — Data cleaning and transformation
- **MySQL** — Business analysis using SQL queries
- **Power BI** — Interactive dashboard and visualization

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**Business Questions Answered (SQL)**

1. What is the total revenue generated by male vs female customers?
2. Which customers used a discount but still spent above average?
3. Which are the top 5 products with the highest average review rating?
4. Compare average purchase amount between Standard and Express Shipping
5. Do subscribed customers spend more than non-subscribers?
6. Which 5 products have the highest discount usage rate?
7. Segment customers into New, Returning, and Loyal
8. What are the top 3 most purchased products within each category?
9. Are repeat buyers more likely to subscribe?
10. What is the revenue contribution of each age group?
11. Which season generates the highest revenue and most orders?

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**Deliverables**

- **Python Notebook** — Cleaned and transformed dataset exported to MySQL
- **SQL Queries** — 11 business queries extracting key insights
- **Power BI Dashboard** — Interactive dashboard with KPIs, charts, and slicers